

NUMBERING SYSTEMS

What is a Numbering System?

- Is a systematic way to represent numbers with symbolic characters and uses a base value to conveniently group numbers in compact form. The most common number system is decimal, which has a base value of 10, and a symbolic character set of 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9.

What is a Numbering System?

- However, there are other number systems, and they can be more efficient to use for a specific purpose. For example, because computers use Boolean logic to perform calculations and operations, they use the binary number system, which has a base value of 2.

Table of Number Systems

Number system	Base value	Symbolic character set
Binary	2	0,1
Octal	8	0, 1, 2, 3, 4, 5, 6, 7
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, and 9
Hexadecimal	16	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

DECIMAL NUMBERING SYSTEMS

DECIMAL

- Most people today use decimal representation to count. In the decimal system there are 10 digits:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Decimal

- The value is formed by the sum of each digit, multiplied by the **base** (in this case it is **10** because there are 10 digits in decimal system) in power of digit position (counting from zero):

Example:

■ 754

$$7 \cdot 10^2 + 5 \cdot 10^1 + 4 \cdot 10^0 = 700 + 50 + 4 = 754$$

Diagram illustrating the expansion of the number 754 into its positional values:

- The base 10 is highlighted in red, with a red line pointing to a box labeled "base".
- The digit position 0 is highlighted in green, with a green line pointing to a box labeled "digit position".

■ 547

$$5 \cdot 10^2 + 4 \cdot 10^1 + 7 \cdot 10^0 = 500 + 40 + 7 = 547$$

Diagram illustrating the expansion of the number 547 into its positional values:

- The base 10 is highlighted in red, with a red line pointing to a box labeled "base".
- The digit position 0 is highlighted in green, with a green line pointing to a box labeled "digit position".

BINARY NUMBERING SYSTEMS

Binary

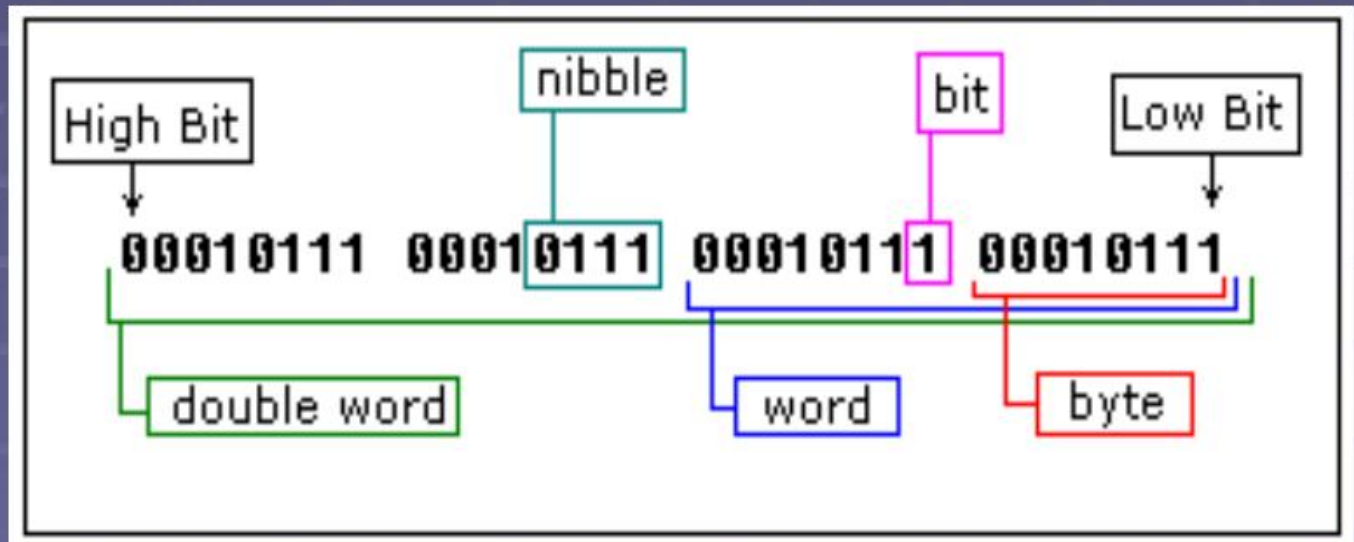
- Computers are not as smart as humans are (or not yet), it's easy to make an electronic machine with two states: **on** and **off**, or **1** and **0**.
Computers use binary system, binary system uses 2 digits:

0, 1

And thus the **base** is **2**.

Binary

- Each digit in a binary number is called a **BIT**, 4 bits form a **NIBBLE**, 8 bits form a **BYTE**, two bytes form a **WORD**, two words form a **DOUBLE WORD** (rarely used):



Example

Word I love you
I = 01101001
space = 00100000
l = 01101100
o = 01101111
v = 01110110
e = 01100101
space = 00100000
y = 01111001
o = 01101111
u = 01110101

Example:

■ 10100101

10100101b =

$$\begin{aligned} &= 1 \cdot 2^7 + 0 \cdot 2^6 + 1 \cdot 2^5 + 0 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 \\ &= 128 + 0 + 32 + 0 + 0 + 4 + 0 + 1 = 165 \end{aligned}$$

(decimal value)

base

digit
position

HEXADECIMAL NUMBERING SYSTEMS

Hexadecimal

- Hexadecimal System uses 16 digits:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

And thus the **base** is **16**.

Hexadecimal

- It is very easy to convert numbers from binary system to hexadecimal system and vice-versa, every nibble (4 bits) can be converted to a hexadecimal digit using this table:

Decimal (base 10)	Binary (base 2)	Hexadecimal (base 16)
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Example:

■ 1234

$$1 \cdot 16^3 + 2 \cdot 16^2 + 3 \cdot 16^1 + 4 \cdot 16^0 = 4096 + 512 + 48 + 4 = 4660$$

(decimal value)

base

digit position

Assignment – $\frac{1}{2}$ CW

- Binary Operations

1). $1101101 + 111$

2). $110001 - 11$

3). $1101101 * 11$

4). $101000 / 10$

Advance Study

- Boolean Algebra
- AND gate
- OR gate
- NOT gate
- NAND
- NOR
- XOR
- XNOR